

Tandem-in-Time Mass Spectrometry with Programmable Clifford Boundary Synthesis

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Abstract

A single analytical electrode cylinder is proposed for tandem mass spectrometry in time. Twenty-four independently weighted blades synthesize a real Clifford boundary schedule so that precursor isolation, activation, fragment capture, recompression, and product-ion analysis are successive states of one intact cage with a detector-facing inter-blade aperture.

Product-ion readout is performed at zero quadrupolar DC by scanning the square-wave $n = 2$ RF amplitude through the first stability boundary while a non-oscillatory detector-directed $n = 1$ field displaces the forced periodic orbit and biases the unstable branch. Unequal RF duty is used as a baseline axis-selectivity refinement. Defining D as the fraction of each period for which the detector-axis coordinate is defocused, $D > 1/2$ lowers the detector-axis instability edge and raises the orthogonal edge. In the reduced model, $D \simeq 0.55$ suppresses off-axis first crossings at substantially smaller DC bias than equal duty while preserving unit-mass separation for the prescribed cooled ensemble.

The inhomogeneous Hill calculation demonstrates directional selection, not instrumental resolving power or absolute transmission. A sufficiently strong DC bias can eliminate opposite-side crossings in the ideal transverse model, but passage through a physical detector still depends on the three-dimensional field, measured waveforms, scan-rate calibration, gas history, cooled phase-space covariance, space charge, and collector optics.

Keywords: tandem-in-time mass spectrometry; birdcage electrode array; digital ion trap; square-wave RF; stability-boundary ejection; Clifford algebra; DC dipole bias

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1 Introduction

Tandem mass spectrometry requires precursor selection, activation, product-ion retention, pressure recovery, and product analysis. Triple quadrupoles distribute these functions among Q1, q2, and Q3, whereas ion traps perform them sequentially in one electrode structure [1, 2, 3, 4]. Linear-trap experiments established mass-selective radial and axial ejection and showed that observed peak shape depends on the ion cloud, auxiliary excitation, scan rate, aperture field, and fringe geometry rather than on an ideal stability boundary alone [5, 6, 7]¹.

Gas-loaded quadrupoles provide the collision-cell precedent: confinement, reaction gas, damping, and ion-molecule chemistry jointly determine transmission and interference removal [8, 9, 10, 11]. Rectangular-wave traps provide the corresponding transfer-matrix and monodromy framework [12, 13, 14, 15, 16, 17, 18]. Multistage tandem operation has already been realized in digital linear and three-dimensional traps by dual-direction dipolar excitation, frequency-scanned and synchronized reverse-scan collision-induced dissociation, and in miniaturized digital instruments [19, 20, 21, 22]. The distinction drawn here is therefore not tandem operation in a digital trap as such. Nor is the general synthesis of static multipole fields by independently biased electrodes itself new. Electrostatic multipole lenses and twelve-electrode dodecapole structures have been developed for deflection, quadrupole focusing, superposed higher-order fields, and aberration correction [23, 24, 25]; related electric and magnetic dodecapole synthesis has also been demonstrated in electron optics [26]. The specific proposal here is to use one 24-blade boundary as a programmable RF/DC analytical volume whose transverse order and temporal role are changed during a tandem-in-time cycle.

The proposed device is a cylindrical array of independently driven blade electrodes. The name *birdcage* is adopted from the NMR birdcage resonator; the analogy is structural, referring to a cylindrical distributed electrode array rather than to the resonant circuit itself [27]. Here the angular weights are imposed electronically, so field order, orientation, and temporal sequence are programmable.

The central readout principle is a separation of functions. During product analysis the quadrupolar DC component is removed, so $a = 0$. The square-wave $n = 2$ RF amplitude is scanned through the first stability boundary and supplies the mass coordinate. A detector-directed $n = 1$ field is superimposed as a DC boundary term. It does not drive a secular resonance; it breaks the transverse sign symmetry and biases the unstable trajectory toward the detector-facing gap. Successive masses reach the same calibrated boundary-crossing region one after another.

The paper extends transverse cylindrical boundary synthesis [28] to a linear trap configuration. It assigns isolation, transverse resonant activation, fragment catch, cooling, and readout to successive states of one cage. Also, it develops detector-directed DC-biased stability-boundary ejection at $a = 0$ as the baseline product-scan example. As a practical solution, it gives first-order constraints on the fragment-catch field, pressure recovery, voltage, timing, directional bias, and validation. The readout treated here is destructive: the product ions are ejected to a detector. A non-destructive image-current terminal stage is a possible alternative but lies outside the scope of the present work².

¹Supplement S3 provides a sequential Q1/q2/Q3 benchmark and discusses the associated programmable-fringe constraints.

²The first-order electrode-load estimates in Supplement S8 are obtained from a symmetry-reduced boundary-element calculation of the present cage geometry.

The intact 24-blade cage is employed as the baseline because it preserves the complete sampled boundary and provides a regular inter-blade detector exit. In the baseline mechanical interpretation, the array is surrounded by a grounded gas-transparent mesh or slotted shield whose mean radius provides the electrostatic reference boundary, with a dedicated detector opening aligned with one selected inter-blade gap. The remaining regular gaps face the surrounding shield or support structure and are not detector channels. The gas-transparent construction is important because the rapid pump-out assumed below requires radial conductance through the inter-blade gaps; a close-fitting solid shield would force the gas predominantly into the much smaller axial conductance.

An open-sector 23-blade variant was evaluated during the design study as a means of accommodating a larger detector entrance close to the cage. Finite-element field calculations showed that removing one blade concentrates the largest field distortion in the detector-facing sector, and direct trajectory modelling consequently produced substantially poorer ejection performance, with a substantially broader event-coordinate distribution than for the intact cage. Because that comparison was a reduced design study rather than an instrumental resolution calculation, no numerical width ratio is used here. The variant was therefore rejected, and the intact 24-blade cage with a regular detector-aligned inter-blade aperture is used throughout. A standing AC dipole is used for baseline precursor activation³.

2 Analyser Architecture and Operating Sequence

The proposed instrument is a single cylindrical analytical cage formed by 24 longitudinally continuous, independently driven blades. Conventional end electrodes provide axial confinement, injection, and clearing, while gas admission and differential pumping establish the collisional and resolving-pressure stages. The distinctive element is the programmable transverse boundary: the same blade array synthesizes the fields required for precursor isolation, resonant activation, fragment retention, collisional recompression, and product-ion mass analysis. The baseline configuration does not use beam-type axial excitation. Because the analytical blades are not axially segmented, no programmable axial acceleration field is synthesized inside the cage; precursor activation is therefore performed transversely. Rectangular-wave precursor isolation in segmented quadrupole structures provides a practical precedent [29]; here the isolation field is synthesized transversely by the continuous 24-blade cage.

The baseline analytical sequence is

- inject and capture $\rightarrow$ isolate precursor in $n = 2$
- $\rightarrow$ cross-fade to $n = 6 + \epsilon_2 n = 2 \rightarrow$ apply resonant AC $n = 1$ under gas
- $\rightarrow$ restore $n = 2$ under gas $\rightarrow$ recompress and cool
- $\rightarrow$ pump to resolving pressure $\rightarrow$ set $a = 0$, $D \simeq 0.55$
- $\rightarrow$ scan $n = 2$ RF with detector-directed DC $n = 1$.

The order of these operations is important. Before activation, the quadrupolar precursor-isolation field is cross-faded to a predominantly $n = 6$ fragment-catch field while retaining a weak residual $n = 2$ component. The $n = 6$ term provides a large-radius confining barrier for

³Its possible reuse for product-ion ejection, together with rotating-field readout, is discussed in Supplement S2.

fragments, whereas the residual quadrupole preserves finite central curvature and a well-defined small-amplitude precursor motion.

Collision-induced dissociation is then driven transversely by a standing AC dipole synthesized on the same blade array,

$$\psi_{1,\text{act}}(t) = V_{\text{act}} \cos(\omega_{\text{act}}t + \phi_{\text{act}})e^{-\sigma_{12}\theta_{\text{act}}}.$$

Here θ_{act} fixes the standing-dipole orientation and ϕ_{act} its temporal phase. The excitation frequency is chosen near the appropriate precursor secular resonance in the instantaneous hybrid catch field. Resonant energy deposition increases the precursor kinetic energy while collisions with the buffer gas convert that driven motion into internal excitation and fragmentation. The fragments are therefore formed while the $n = 6$ catch is already present and do not first encounter the low-mass cutoff of the final quadrupolar analysis field.

The activation amplitude V_{act} , frequency or short frequency law $\omega_{\text{act}}(t)$, phase ϕ_{act} , duration t_{act} , gas pressure, and retained quadrupole fraction ϵ_2 are experimental control parameters. Their optimum values depend on precursor mass, collision gas, desired fragmentation, and collisional damping and must ultimately be established experimentally or by collisional trajectory calculation. The present work therefore specifies the activation architecture rather than assigning a universal collision energy to the resonantly driven packet.

After activation, the AC $n = 1$ drive is removed and the $n = 2$ field is restored while damping gas remains present. The packet is held for several damping times so that activation-induced coherent motion and the $n = 6 \rightarrow n = 2$ transition decay before pump-out. Product-ion analysis then uses a distinct, non-oscillatory detector-directed $n = 1$ coefficient: at $a = 0$ and unequal RF duty, the $n = 2$ amplitude is scanned through the first stability boundary while the DC $n = 1$ field selects the detector-facing branch of the instability.

Table 1 summarizes the principal control elements and their time-ordered roles. The AC and DC $n = 1$ entries use the same spatial dipole basis but are temporally distinct operations.

Table 1: Principal control elements and operating modes of the baseline 24-blade tandem-in-time analyser.

Control element	Baseline function	Principal stage
End electrodes	Axial confinement, injection, and clearing; no baseline CID excitation	Injection and storage
AC $n = 1$	Standing transverse secular excitation of the precursor under collision gas	CID activation
$n = 2$ RF	Precursor isolation, residual central curvature during activation, recompression, and scanned product-ion mass coordinate	Isolation, activation support, cooling, analysis
$n = 6$ RF	Large-radius fragment catch established before and maintained during CID	Activation and fragment retention
DC $n = 1$	Detector-directed displacement and branch selection during stability-boundary ejection	Product-ion readout

The angular modes are synthesized from the independently driven blade voltages using the cylindrical boundary formalism developed in Section 3. The mathematical synthesis supplies

a common control framework; the analytical function of the device is determined by the time-ordered sequence of the physical operating modes above.

A front accumulation trap may fill while the analytical cage processes the preceding packet. If T_{acc} is the accumulation time and $T_{\text{analytical}}$ is the duration of one complete analytical sequence, the effective packet period is approximately $T_{\text{packet}} \simeq \max(T_{\text{acc}}, T_{\text{analytical}})$. Parallel accumulation is therefore an important throughput feature, although it is not required for the underlying single-cage tandem-in-time principle. Timing estimates are given in Table 6⁴.

3 Clifford Boundary and 24-Blade Synthesis

We work in the real spacetime algebra $\text{Cl}(1, 3)$ of Hestenes [30],

$$e_\mu e_\nu + e_\nu e_\mu = 2\eta_{\mu\nu}, \quad \eta_{\mu\nu} = \text{diag}(+1, -1, -1, -1).$$

The relative spatial vectors $\sigma_k = e_k e_0 \in \text{Cl}^+(1, 3)$ satisfy

$$\sigma_j \sigma_k + \sigma_k \sigma_j = 2\delta_{jk}.$$

Define

$$\mathbf{r}_\perp = x\sigma_1 + y\sigma_2, \quad \zeta = \sigma_1 \mathbf{r}_\perp = x + y\sigma_{12} = \rho e^{\sigma_{12}\theta},$$

where $\sigma_{12} = \sigma_1 \sigma_2$ and $\sigma_{12}^2 = -1$. The transverse phase subalgebra is

$$\mathcal{C}_\perp = \text{span}_{\mathbb{R}}\{1, \sigma_{12}\} \subset \text{Cl}^+(1, 3).$$

Every transverse phase and sine-cosine pair used below lies in $\mathcal{C}_\perp$. Because $\mathcal{C}_\perp$ is commutative, factors drawn from it may be reordered freely inside a scalar-grade projection $\langle \cdot \rangle_0$; this property is used without further comment. Real transfer, finite-element, and circuit matrices are numerical operators over $\mathbb{R}$; they do not introduce another physical algebra.

Transverse synthesis. The programmable boundary on the reference cylinder $\rho = R$ is the real schedule

$$g(\theta, t) = \sum_{n=0}^{12} \langle \psi_n(t) e^{n\sigma_{12}\theta} \rangle_0, \quad \psi_0(t) \in \mathbb{R}, \quad \psi_n(t) \in \mathcal{C}_\perp \quad (1 \leq n \leq 12).$$

For $n \geq 1$,

$$\psi_n(t) = A_n(t) e^{-n\sigma_{12}\varphi_n(t)}$$

encodes amplitude and orientation, so that $\langle \psi_n e^{n\sigma_{12}\theta} \rangle_0 = A_n \cos n(\theta - \varphi_n)$. For a boundary that does not vary along the axis, the interior Dirichlet solution is

$$\Phi_n(\rho, \theta, t) = \left(\frac{\rho}{R}\right)^n \langle \psi_n(t) e^{n\sigma_{12}\theta} \rangle_0, \quad n \geq 1, \quad (1)$$

which is the axially uniform limit of the general cylindrical solution developed in [28]. The corresponding electric field $\mathcal{E}_n = -\nabla_\perp \Phi_n$ is a relative vector in $\text{Cl}^+(1, 3)$; the multipole identity that produces it is given in Appendix A. In the baseline analyser the 24 blades are longitudinally

⁴The electronics required to realize the programmed boundary sequence are discussed in Supplement S8.

continuous, so every programmed coefficient is independent of the axial coordinate s over the active central length and Eq. (1) is the working synthesis formula of this paper. The baseline therefore has no programmable axial channel: the $n = 0$ coefficient is an s -independent common-mode offset, and axial confinement, injection, and clearing are supplied by the end electrodes. A deliberately s -dependent schedule would require separate end electrodes or axially segmented blade groups; such travelling structures are a possible extension [31] but are not used here.

Blade sampling. For 24 blade centres,

$$\theta_k = \theta_{\text{ref}} + \frac{2\pi k}{24}, \quad k = 0, \dots, 23,$$

the commanded blade voltage is

$$V_k(t) = \sum_{n=0}^{12} W_n \langle \psi_n(t) e^{n\sigma_{12}\theta_k} \rangle_0, \quad W_n = \frac{1}{\text{sinc}(n\Delta\theta_b/2)}, \quad (2)$$

where $\text{sinc } x \equiv \sin x/x$, with $\text{sinc } 0 = 1$. The illustrative baseline uses the blade fill factor

$$d_b = \frac{\Delta\theta_b}{2\pi/24} = \frac{1}{2}, \quad \Delta\theta_b = \Delta\theta_g = \frac{\pi}{24},$$

so each blade and each regular gap subtend 7.5° . The finite-width factors are

$$W_1 = 1.0007, \quad W_2 = 1.0029, \quad W_6 = 1.0262, \quad W_{12} = 1.1107.$$

These values use $\Delta\theta_b = \pi/24$; in particular, $W_2 = 1/\text{sinc}(\pi/24)$. On the 24-site lattice $e^{12\sigma_{12}\theta_k} = (-1)^k e^{12\sigma_{12}\theta_{\text{ref}}}$, so the $n = 12$ Nyquist channel contributes only the single real alternating pattern $A_{12} \cos 12(\theta_{\text{ref}} - \varphi_{12}) (-1)^k$. Counting one real ψ_0 , two real components for each of $n = 1, \dots, 11$, and one for $n = 12$ gives $1 + 2(11) + 1 = 24$, matching the number of blades.

The sampled alias family is

$$m \in \{|24\ell \pm n| : \ell = 0, 1, 2, \dots\}. \quad (3)$$

For the $d_b = 1/2$ baseline, the leading quadrupole alias contributes less than 8×10^{-6} of the quadrupole field at $\rho = R/2$; the leading $n = 6$ alias contributes approximately 5.9×10^{-4} there⁵.

Commanded coefficient and realized field. The factor W_n corrects the harmonic *shape* of the sampled boundary; it does not fix the absolute amplitude of the interior field, which depends on the mechanical realization. Two levels of description are used in this paper. The arc-sector boundary model of this section and of Supplement S1 treats each blade as a conducting arc of width $\Delta\theta_b$ lying on $\rho = R$, with the gaps at the grounded reference; it is the synthesis abstraction used for the alias estimates above. The mechanical realization analysed in Supplement S1 (Fig. S1) and Supplement S8 is a set of thin radial blades whose inner tips lie on $R_0 = R = 4$ mm and which extend outward inside a grounded gas-transparent shield. For that geometry the boundary-element solution gives a quadrupole coefficient of 0.869 of the peak blade voltage at R_0 , with alias content well below the arc-sector estimate. To keep the two descriptions consistent, write

$$V_n^{\text{blade}}(t) = \frac{W_n}{\kappa_n} V_n(t), \quad \kappa_2 = 0.869 \text{ (radial-blade geometry of Fig. S1)}, \quad (4)$$

⁵Supplement S1 gives the blade-fill-factor dependence.

where V_n is the realized interior coefficient of order n referred to $\rho = R$, V_n^{blade} is the corresponding peak blade voltage, and κ_n is the geometric coupling of that order. Throughout the paper the symbols V_{rf} , V_d , V_n , and V_{act} denote realized interior coefficients, as required by the definitions of q , q_d , and q_n in Sections 4 and 5; rail and blade voltages are obtained from them through Eq. (4). Only κ_2 has been solved for the present geometry; κ_1 and κ_6 must be obtained from the same boundary-element model before the corresponding rail voltages are fixed.

For finite-length fringe analysis, and for possible future axially segmented variants, the coefficients acquire an axial dependence $\psi_n(s, t) \in \mathcal{C}_\perp$. With the axial Clifford-Fourier transform

$$\widehat{\psi}_n(k, t) = \int_{-\infty}^{\infty} \psi_n(s, t) e^{-\sigma_{12}ks} ds \in \mathcal{C}_\perp,$$

the ideal cylindrical Dirichlet solution for $n \geq 1$ is

$$\Phi_n(\rho, \theta, s, t) = \int_{-\infty}^{\infty} \frac{dk}{2\pi} \frac{I_n(|k|\rho)}{I_n(|k|R)} \left\langle \widehat{\psi}_n(k, t) e^{\sigma_{12}(n\theta+ks)} \right\rangle_0, \quad (5)$$

with the analogous I_0 factor for $n = 0$. As $k \rightarrow 0$ the Bessel kernel reduces to $(\rho/R)^n$ and Eq. (5) reduces to Eq. (1); the derivation is given in Appendix B⁶.

4 Square-Wave Quadrupole Stability Boundary

Let the ion charge be $Q = \pm ze$, where $z > 0$ is the charge-number magnitude. Define the positive SI mass-to-charge ratio $\mu = m/(ze)$, and let $(m/z)_{\text{Th}}$ denote its numerical value in thomson:

$$\mu = \frac{u}{e} (m/z)_{\text{Th}}.$$

The stability parameters below use $\mu > 0$; the electrode polarity is reversed for negative ions. The square-wave (digital) operation of the quadrupole follows established rectangular-wave practice [12, 13, 17]. For the equal-duty bipolar waveform $\chi = \pm 1$,

$$\Phi_2(x, y, t) = (U - V_{\text{rf}}\chi(t)) \frac{x^2 - y^2}{R^2}.$$

With $\tau = \Omega t/2$,

$$\begin{aligned} x'' + (a - 2q\chi(\tau))x &= 0, \\ y'' + (-a + 2q\chi(\tau))y &= 0, \end{aligned}$$

where

$$a = \frac{8U}{\mu R^2 \Omega^2}, \quad q = \frac{4V_{\text{rf}}}{\mu R^2 \Omega^2}.$$

The voltage convention is that V_{rf} is the zero-to-peak pole-to-centre amplitude of the realized quadrupole coefficient at $\rho = R$, and

$$V_{\text{pp,rail}} = \frac{2V_{\text{rf}}}{\kappa_2}, \quad V_{\text{pp,diff}} = \frac{4V_{\text{rf}}}{\kappa_2},$$

⁶In the baseline configuration Eq. (5) enters only through the finite-blade-length and end-field attenuation discussed in Supplement S3.

with κ_2 the geometric coupling of Eq. (4) ($\kappa_2 = 0.869$ for the radial-blade geometry; $\kappa_2 = 1$ for a coefficient referred directly to the blade voltage)⁷.

The equal-duty stability constants follow from rectangular-wave monodromy [32, 33, 14]:

$$a_{\text{tip}} = 0.238110455, \quad q_{\text{tip}} = 0.552873144, \quad \frac{a_{\text{tip}}}{q_{\text{tip}}} = 0.430678281,$$

and the first RF-only boundary at $a = 0$ is

$$q_b = q_{\text{RF,max}} = 0.712493657 = \frac{2\alpha_1^2}{\pi^2}, \quad \cosh \alpha_1 \cos \alpha_1 = -1.$$

The compact transfer formula is given in Appendix C.

At $a = 0$, the secular ratio satisfies

$$\cos(\pi\beta) = \frac{1}{2} \text{Tr } M(q) = \cos \alpha \cosh \alpha, \quad \alpha = \frac{\pi}{2} \sqrt{2q},$$

where $\beta = 2\omega_0/\Omega$. At the ideal first boundary,

$$q = q_b \implies \text{Tr } M = -2, \quad \beta_b = 1.$$

During product readout, $U = 0$, so $a = 0$. The mathematical boundary is q_b , but a finite scan rate and the inhomogeneous DC bias produce a calibrated first-crossing coordinate q_e near q_b . Mass reconstruction must use the measured q_e , not assume an exact static boundary crossing.

For $(m/z)_{\text{Th}} = 1000$, $R = 4$ mm, and $f_{\text{RF}} = 1$ MHz, the ideal $a = 0$ boundary requires a realized quadrupole coefficient of approximately $V_{\text{rf}} = 1166$ V. For the radial-blade geometry with $\kappa_2 = 0.869$ (Eq. (4)) this corresponds to a peak blade voltage of $V_{\text{rf}}/\kappa_2 \simeq 1342$ V and $V_{\text{pp,rail}} = 2V_{\text{rf}}/\kappa_2 \simeq 2684$ V.

5 Detector-Directed DC Bias and Stability-Boundary Ejection

Let the detector gap be centred at azimuth θ_g . For positive ions, choose the DC $n = 1$ potential

$$\Phi_{1,\text{dc}}(\rho, \theta, t) = -V_d(t) \frac{\rho}{R} \cos(\theta - \theta_g),$$

so that its electric field points toward the detector. In the Clifford boundary representation,

$$\psi_{1,\text{dc}}(t) = -V_d(t) e^{-\sigma_{12}\theta_g} \in \mathcal{C}_\perp.$$

The sign is reversed for negative ions. The blade voltage during readout is

$$V_k(t) = -\frac{W_2}{\kappa_2} V_{\text{rf}}(t) \chi(t) \cos 2(\theta_k - \theta_g) - \frac{W_1}{\kappa_1} V_d(t) \cos(\theta_k - \theta_g),$$

with κ_1, κ_2 the geometric couplings of Eq. (4). The $n = 1$ term is non-oscillatory. It is continuously present during the RF-amplitude scan and is not a resonant auxiliary excitation.

Fig. 1 collects ideal transverse channel topologies. Panel (e) is the $n = 2$ quadrupole rotated by 7.5° , half the 15° blade pitch, illustrating that the field orientation φ_2 is set electronically

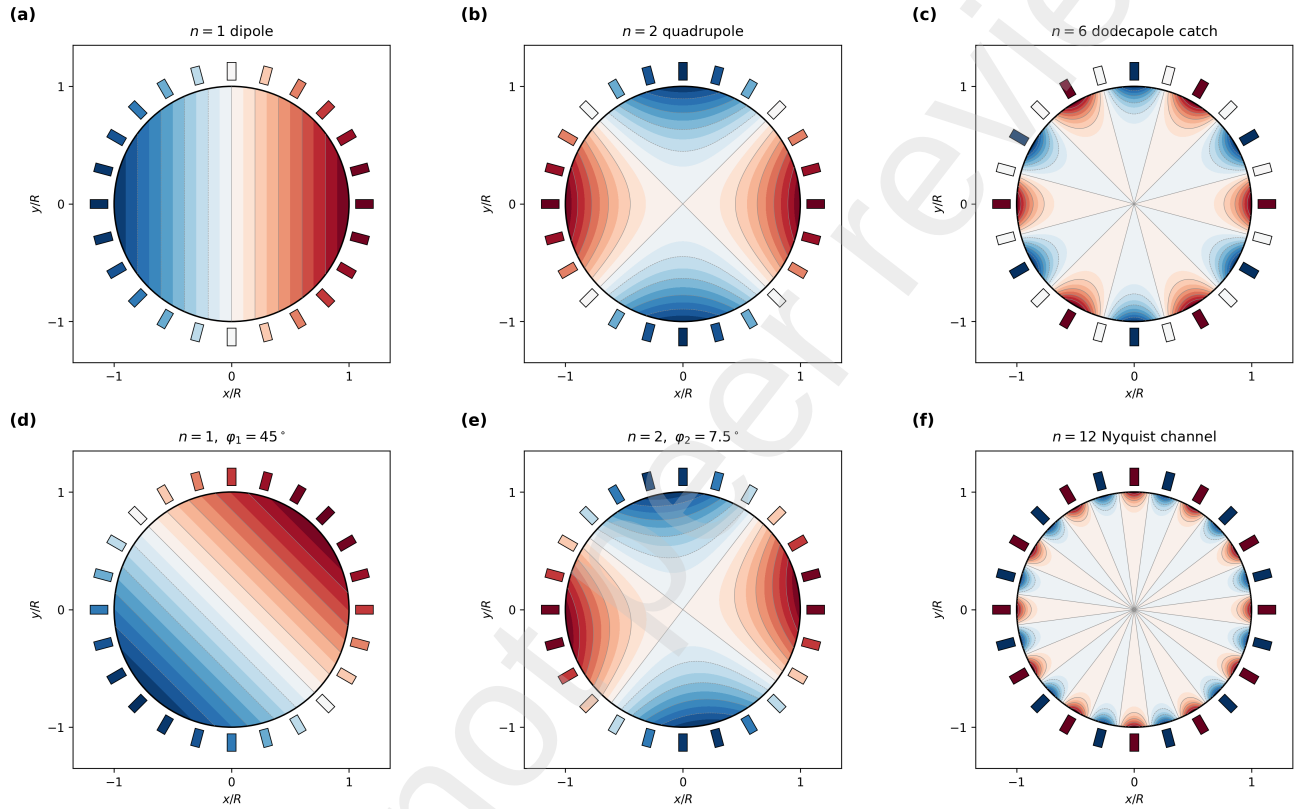


Figure 1: The palette of ideal transverse channels synthesized on the intact 24-blade boundary. Blue denotes negative potential or blade voltage, white denotes zero, and red denotes positive potential or blade voltage. In every panel the electrode colors are calculated from the sampled boundary function that generates the interior field. The six panels show (a) the detector-aligned $n = 1$ dipole, (b) the $n = 2$ quadrupole, (c) the $n = 6$ dodecapole catch, (d) an $n = 1$ dipole rotated to $\varphi_1 = 45^\circ$, (e) an $n = 2$ quadrupole rotated to $\varphi_2 = 7.5^\circ$, and (f) the $n = 12$ Nyquist channel.

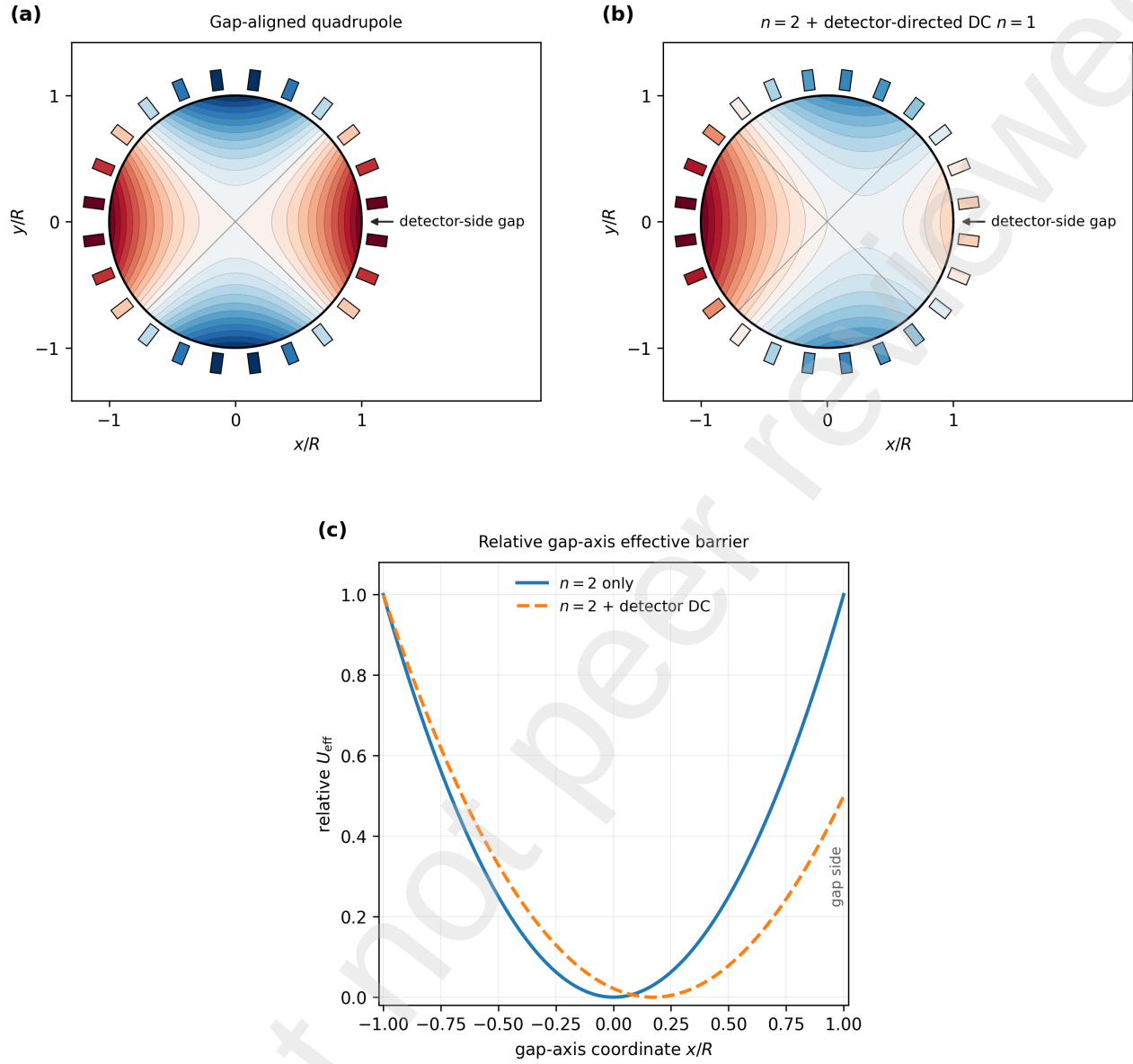


Figure 2: The detector-directed ejection field itself, on the 24-blade cage with a regular 7.5° inter-blade gap centred on the detector axis, showing the gap-aligned quadrupole, its superposition with the detector-directed DC term, and the resulting barrier asymmetry. The selected gap is assumed to align with the dedicated detector opening in the surrounding grounded gas-transparent shield, which is not shown. In the signed maps, blue is negative, white is zero, and red is positive; electrode colors are the corresponding sampled boundary voltages. (a) Gap-aligned quadrupole. (b) Instantaneous quadrupole potential plus a detector-directed DC $n = 1$ term. (c) Qualitative reduced gap-axis pseudopotential illustration: the DC term shifts the apparent well toward the detector and lowers the detector-side barrier. The small-micromotion approximation is not quantitatively controlled near the first stability boundary, so this panel is not used to predict the ejection threshold. Optimized higher odd DC terms are not included. The pseudopotential is used only for this qualitative illustration.

through the blade weights rather than pinned to the electrode positions (the same freedom that aligns the detector-directed $n = 1$ axis with the selected gap). Then Fig. 2 shows the detector-side geometry. The qualitative gap-axis pseudopotential used in Fig. 2(c) is defined in Appendix D.

Let x_g be the transverse coordinate directed from the cage axis toward the detector-gap centre, and let y_g be the orthogonal transverse coordinate. With

$$X = \frac{x_g}{R}, \quad Y = \frac{y_g}{R}, \quad \tau = \frac{\Omega t}{2},$$

the ideal transverse equations are

$$X_i'' - 2q_i(\tau)\chi(\tau)X_i = q_{d,i}(\tau), \quad (6)$$

$$Y_i'' + 2q_i(\tau)\chi(\tau)Y_i = 0, \quad (7)$$

where

$$q_i(t) = \frac{4V_{\text{rf}}(t)}{\mu_i R^2 \Omega^2}, \quad q_{d,i}(t) = \frac{4V_d(t)}{\mu_i R^2 \Omega^2}.$$

The DC term makes Eq. (6) inhomogeneous, but it does not change the homogeneous monodromy matrix or the boundary q_b . Over one RF period, writing $\mathbf{w}_j = (X_j, X'_j)^T$, the detector-axis state obeys the affine map

$$\mathbf{w}_{j+1} = M(q)\mathbf{w}_j + q_d \mathbf{b}(q), \quad (8)$$

and the stable forced periodic state is

$$\mathbf{w}_c(q, q_d) = [I - M(q)]^{-1} q_d \mathbf{b}(q). \quad (9)$$

As the RF amplitude is scanned, this forced orbit moves. Once the homogeneous motion becomes dynamically unstable, the DC displacement gives the growing solution a preferred detector-side projection.

Away from the stability edge, the secular estimate is

$$\frac{x_c}{R} \simeq \frac{q_d}{\beta^2}.$$

This estimate explains the directional displacement but is not used as an exact near-boundary transfer law. An oscillation about the displaced centre reaches the detector-side wall before the opposite wall when

$$R - x_c \leq A_x < R + x_c.$$

Thus opposite-side loss is not unavoidable. Deliberate transverse asymmetry is an established route to directional ejection, whether imposed geometrically through the electrode profile [34] or, as here, electronically through the boundary coefficients. However, both transverse homogeneous modes approach the same equal-duty stability boundary at $a = 0$. A DC bias along x_g suppresses the opposite branch but does not automatically suppress Y -growth or crossings outside a narrow detector sector. This limitation motivates the higher-odd-order DC shaping considered below.

⁷Supplement S5 summarizes this voltage convention and the DC-biased boundary-scan implementation.

Cooled initial ensemble. The first-crossing distribution depends strongly on the ion phase-space distribution at the start of the product scan. After collisional activation and fragment capture, the ions should not be transferred abruptly from the flat central region of a predominantly $n = 6$ field into the final collision-free $n = 2$ scan. The preferred sequence is

$$(\Phi_6 + \epsilon_2 \Phi_2) + \text{gas} \longrightarrow [A_6(t)\Phi_6 + A_2(t)\Phi_2] + \text{gas} \longrightarrow \Phi_2 + \text{gas} \longrightarrow (\Phi_2 + \Phi_{\text{steer}}) + \text{gas},$$

followed by pump-out and the RF-amplitude scan. Here $0 < \epsilon_2 \ll 1$ is the residual quadrupole amplitude fraction during fragment capture, and $A_6(t)$ and $A_2(t)$ implement the cross-fade.

Restoring $n = 2$ introduces finite central curvature and narrows the thermal spatial distribution. For an approximately harmonic secular mode with effective transverse temperature $T_\perp$, mass m , and secular angular frequencies ω_x and ω_y ,

$$\sigma_x \simeq \sqrt{\frac{k_B T_\perp}{m \omega_x^2}}, \quad \sigma_y \simeq \sqrt{\frac{k_B T_\perp}{m \omega_y^2}}.$$

The $n = 2$ field should therefore be restored while damping gas is still present and held for several damping times before pump-out. A sudden collision-free switch would instead excite coherent breathing and sloshing.

In the exact square-wave field, the cooled packet is concentrated around an RF-phase-dependent periodic orbit, not a motionless point in a static pseudopotential. With the DC steering field present, that orbit is $\mathbf{w}_c(q, q_d)$ from Eq. (9). A physically complete simulation must obtain the phase-dependent covariance around this orbit from the collisional $n = 6 \rightarrow n = 2$ transfer rather than prescribe independent coordinate and velocity widths.

Detector-side DC shaping. Because the $n = 1$ boundary coefficient is synthesized independently of the RF $n = 2$ coefficient, its direction is not blurred by the quadrupole. A pure detector-aligned $n = 1$ potential,

$$\Phi_{1,\text{dc}} = -V_1 \frac{\rho}{R} \cos(\theta - \theta_g),$$

produces the uniform transverse field

$$\mathbf{E}_{1,\text{dc}} = \frac{V_1}{R} \hat{\mathbf{e}}_g,$$

where $\hat{\mathbf{e}}_g = \sigma_1 e^{i\sigma_{12}\theta_g} = \cos \theta_g \sigma_1 + \sin \theta_g \sigma_2 \in \text{Cl}^+(1, 3)$ is the unit relative vector pointing toward the detector; in the same notation the detector-frame coordinates satisfy $x_g + y_g \sigma_{12} = \zeta e^{-i\sigma_{12}\theta_g}$. Thus pure $n = 1$ gives an exact global direction. It does not, however, create a narrow angular beam at the aperture.

A proposed near-wall refinement is to add weak higher odd DC orders:

$$\Phi_{\text{steer}}(\rho, \theta) = - \sum_{p=0}^5 V_{2p+1} \left(\frac{\rho}{R} \right)^{2p+1} \cos[(2p+1)(\theta - \theta_g)]. \quad (10)$$

The $n = 1$ term supplies the global displacement, whereas the fields of $n = 3, 5, \dots, 11$ scale as

$$|\mathbf{E}_n| \propto \frac{n|V_n|}{R} \left(\frac{\rho}{R} \right)^{n-1}$$

and are increasingly concentrated near the wall. Restricting the refinement to odd orders avoids a direct quadrupolar DC coefficient in the central expansion. This is a proposed optimization variable, not a demonstrated extraction solution: the coefficients require a three-dimensional aperture model, voltage and core-field constraints, and recalibration of q_e . The reduced calculation of Fig. 3 uses only the pure detector-directed $n = 1$ bias.

For a fixed physical V_d , $q_d \propto 1/\mu$. This is adequate over a narrow product window. Over a broad scan, a slowly varying DC law

$$V_d(t) = \kappa V_{\text{rf}}(t) \quad (11)$$

keeps $q_d/q = \kappa$ constant for every mass. Equation (11) is still a DC-only $n = 1$ field: it follows the slow scan envelope and contains no RF or secular-frequency oscillation. Keeping q_d/q constant does not, however, make the dynamic ejection coordinate mass independent. For a linear voltage ramp,

$$\dot{q}_i = \frac{4\dot{V}_{\text{rf}}}{\mu_i R^2 \Omega^2},$$

so different masses encounter the boundary at different dimensionless scan rates. A broad scan therefore requires either a calibrated $q_e(\mu, \dot{q}, \kappa)$ or a scan law designed to keep the relevant dimensionless rate approximately constant, for example a nearly constant fractional voltage rate $d \ln V_{\text{rf}}/dt$.

For an accepted detector crossing j at time $t_{e,j}$, define the instantaneous crossing coordinate $q_j = q_i(t_{e,j})$. The assigned mass coordinate uses a calibrated central crossing value q_e :

$$\left(\widehat{m/z}\right)_{j,\text{Th}} = \frac{e}{u} \frac{4V_{\text{rf}}(t_{e,j})}{q_e R^2 \Omega^2}. \quad (12)$$

Individual events need not cross at exactly q_e ; their distribution about q_e produces the reconstructed peak width. The calibrated value is not assumed universal and can depend on mass coordinate, DC bias, scan law, initial distribution, and aperture geometry.

A reduced ideal-field calculation was performed using Eqs. (6) and (7). To make the recalculation reproducible, the reference coordinate for $m/z = 501$ is scanned linearly from $q_{\text{ref}} = 0.660$ to 0.730 over 1600 RF periods, with four exact constant-segment propagation substeps per waveform segment. For the three-mass comparison, 250 ions of each of $m/z = 500$, 501, and 502 are launched from a diagonal Gaussian ensemble about the phase-matched forced periodic orbit with normalized widths $\sigma_X = \sigma_{X'} = \sigma_Y = \sigma_{Y'} = 10^{-4}$. The detector sector is 7.5° wide. A fixed physical detector-directed bias is used over this narrow mass interval, so the dimensionless $q_{d,i}$ scales as $501/(m/z)_i$. The prescribed Gaussian ensemble is a controlled numerical benchmark; it was not generated by a collisional $n = 6 \rightarrow n = 2$ transfer and therefore is not a validated post-cooling covariance.

At the illustrative equal-duty bias $q_d = 0.20$ for the 501-Th reference mass, the uniform detector-directed $n = 1$ field eliminates opposite-side first crossings in this benchmark but does not suppress every off-axis trajectory. Small initial projections on the orthogonal Y mode can grow during the RF scan and produce two non-detector-sector branches near $\theta_{\text{cross}} \simeq \pm 70^\circ$. Increasing the uniform $n = 1$ bias reduces these branches, but excessive uniform bias also displaces the forced periodic orbit toward the wall and can produce premature ejection before the intended mass reaches the calibrated stability-boundary region. Increasing the uniform $n = 1$ voltage is therefore not, by itself, a sufficient design rule. The higher-odd-order field of Eq. (10)

is retained only as a candidate variable for the subsequent three-dimensional optimization; no transmission improvement from that field is claimed by the present calculation.

The reduced calculation terminates trajectories at the ideal circular reference boundary $\rho = R$. Its “detector”, “opposite”, and “other” categories are angular first-crossing categories, not a resolved classification of metal impacts or passage through a physical port. Only the selected inter-blade gap is aligned with the detector opening in the gas-transparent shield. In the plotted labels, “other azimuth” denotes any crossing outside the selected detector sector. The subsequent fate of such a trajectory depends on the actual blade, shield, support, aperture, and collector geometry and requires the three-dimensional model.

For this recalculation, the equal-duty single calibration coordinate obtained from the accepted $m/z = 501$ events is $q_e = 0.714821$. It is applied unchanged to the $m/z = 500$ and 502 events, with no three-point affine correction⁸. All 250 trajectories per mass reach the ideal reference boundary before the programmed end of this scan. The outcome counts and accepted-event statistics are reported in Table 2.

Fig. 3 shows that, within this reduced model, increasing the detector-directed DC term suppresses the opposite-side branch and concentrates the surviving first crossings into the detector-facing sector, while a residual fraction still reaches other angular sectors of the ideal reference boundary. The reduced model does not determine whether these crossings strike metal or pass through another physical gap. The reconstructed accepted-event spectrum is therefore an illustration of directional selection in the reduced transverse model, not a full instrument prediction. The result remains sensitive to the cooled phase-space covariance, the $n = 6 \rightarrow n = 2$ recompression history, the DC ramp, and the near-gap field. Direct collisional-transfer and concentrated-DC trajectory calculations are required before assigning a detector transmission.

Table 2: Outcome counts and accepted-event statistics from the recalculated equal-duty benchmark of Fig. 3. Each mass has $N_{\text{launch}} = 250$, $q_d = 0.20$ for the 501-Th reference mass, and $q_e = 0.714821$. The reported standard deviations describe the numerical ensemble only and are not instrumental resolving powers.

True m/z	N_{det}	N_{other}	N_{opp}	$N_{\text{uncrossed}}$	Mean $\widehat{m/z}$	SD (Th)
500	169	81	0	0	500.267	0.139
501	160	90	0	0	501.000	0.126
502	167	83	0	0	501.961	0.145

Table 2 reports all launched trajectories. Let N_{det} , N_{other} , N_{opp} , and $N_{\text{uncrossed}}$ denote, respectively, detector-sector, other-azimuth, opposite-side, and no-crossing outcomes, so that $N_{\text{launch}} = N_{\text{det}} + N_{\text{other}} + N_{\text{opp}} + N_{\text{uncrossed}}$. For this explicit slow-scan benchmark $N_{\text{uncrossed}} = 0$ for all three masses. The detector fractions are therefore first-crossing angular fractions on the ideal circular boundary, not physical detector transmissions. Across 500–502 Th the mass-dependent scan rates differ only slightly; small changes in RF phase and first-crossing coordinate redistribute events between the detector sector and the off-axis branches.

Let D denote the fraction of each RF period during which the detector-axis coordinate X is defocused. For the zero-mean waveform

$$\tilde{\chi}_D(t) = \chi_D(t) - (2D - 1),$$

⁸see Supplement S7.

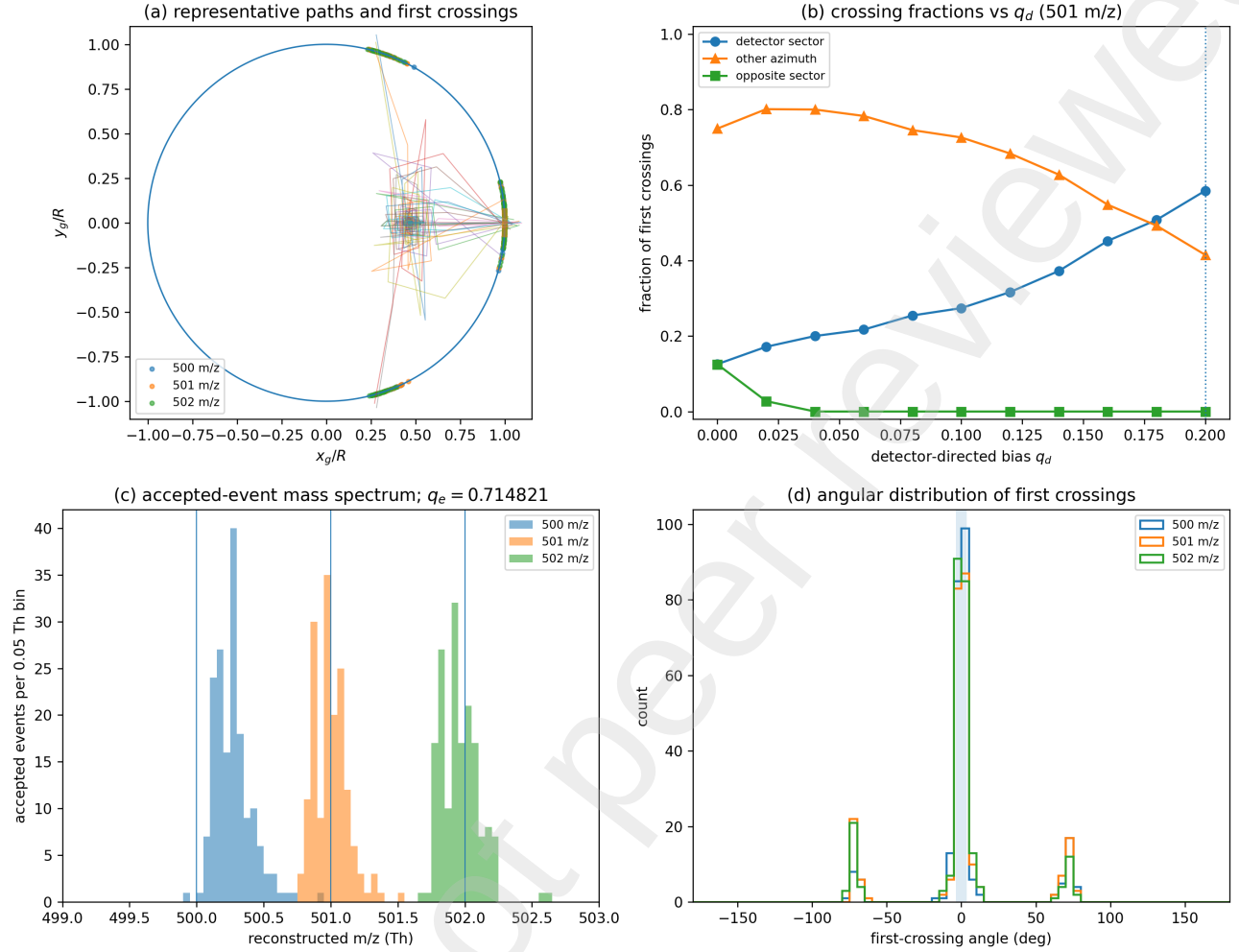


Figure 3: Recalculated reduced two-dimensional DC-biased stability-boundary model (intra-period micromotion; equal duty, $a = 0$). The cooled benchmark uses a 501-Th reference scan $q_{\text{ref}} = 0.660 \rightarrow 0.730$ over 1600 RF periods and the explicit diagonal normalized Gaussian widths stated in the text. (a) Representative paths and all first boundary crossings at $q_d = 0.20$; the detector-facing 7.5° sector is marked. (b) First-crossing fractions versus detector-directed bias for $m/z = 501$; 2000 ions are used at each bias value. The opposite-side fraction falls to zero by $q_d \simeq 0.04$, while an off-axis branch remains at larger bias. (c) Accepted-event mass spectrum at $q_d = 0.20$, reconstructed through Eq. (12) with the single 501-Th calibration $q_e = 0.714821$, applied unchanged to the 500- and 502-Th events. No three-point affine calibration, detector-response convolution, or smoothing is used. (d) Angular distribution of all first crossings for the same 250-ion three-mass calculation. The model is ideal and transverse: it excludes the three-dimensional aperture field, finite blade length, collisions, space charge, end fields, and collector optics. The numerical peak widths therefore characterize only this prescribed ensemble and are not instrumental resolving powers.

the level $+2(1-D)$ is applied for fraction D , and the level $-2D$ for the remainder. For $D \neq 1/2$, V_{rf} therefore denotes one half of the fixed peak-to-peak quadrupole-coefficient swing: the two physical coefficient levels are $+2(1-D)V_{\text{rf}}$ and $-2DV_{\text{rf}}$. Equal duty recovers the symmetric levels $\pm V_{\text{rf}}$. Maintaining $a = 0$ at unequal duty requires these asymmetric zero-mean levels (or an exactly compensating quadrupolar DC offset); simply changing the dwell times of symmetric $\pm V_{\text{rf}}$ rails would introduce a nonzero mean quadrupole component. Because the detector-axis equation is $X'' - 2q\tilde{\chi}_D X = q_d$, the positive segment is the detector-axis-defocusing segment. Thus $D > 1/2$, not $D < 1/2$, lowers the detector-axis instability edge and raises the orthogonal-mode edge. At $D = 1/2$ the two transverse equations are related by a half-period shift; at unequal duty the edges split, with the symmetry $q_{\text{edge}}^{(x)}(D) = q_{\text{edge}}^{(y)}(1-D)$. Table 3 gives the computed edges. Variation of the trapping duty cycle is an established experimental handle on the digital stability diagram [35, 15]; what is used here is the resulting splitting of the two transverse edges rather than the displacement of the diagram as a whole. Duty is a global waveform parameter, not a blade-by-blade multipole coordinate, and its use requires calibration of the dynamic crossing coordinate under the chosen scan law.

Table 3: First transverse instability edges at $a = 0$ for a zero-mean asymmetric rectangular wave at fixed peak-to-peak amplitude. The signed split is $[q_{\text{edge}}(x) - q_{\text{edge}}(y)]/q_{\text{edge}}(y)$.

D	$q_{\text{edge}}(x)$	$q_{\text{edge}}(y)$	split
0.35	0.847448	0.731168	15.9%
0.40	0.781285	0.708373	10.3%
0.45	0.737923	0.702735	5.0%
0.48	0.720706	0.706767	2.0%
0.50	0.712494	0.712494	0.0%
0.52	0.706767	0.720706	-1.9%
0.55	0.702735	0.737923	-4.8%
0.60	0.708373	0.781285	-9.3%
0.65	0.731168	0.847448	-13.7%

This unequal-duty refinement relaxes both the residual off-axis loss and the bias limitation noted above. At equal duty with $a = 0$, the detector-axis and orthogonal modes reach the stability boundary together, so directional selection depends entirely on the DC displacement. At $D \simeq 0.55$, the scanned mass becomes unstable first along the detector axis while the orthogonal mode remains confined over a finite interval. The recalculated diagnostic sweep in Fig. 4 shows a broad high-selection region for approximately $0.53 \lesssim D \lesssim 0.60$ rather than a sharply defined optimum. The value $D = 0.55$ is retained as the baseline compromise because Table 3 gives a 4.8% signed separation of the transverse edges there while the detector-axis edge remains close to its minimum value.

For the separately recalculated cooled three-mass ensemble at $D = 0.55$ and $q_d = 0.10$, the 501-Th events (1000 ions per mass) give their own dynamic calibration $q_e = 0.705869$, rather than the equal-duty value. Applying this one calibration unchanged to 500 and 502 Th preserves unit-mass separation in the numerical event coordinate; the outcome counts and accepted-event statistics are reported in Table 4, in the same format as Table 2. All 3000 launched trajectories reach the ideal reference boundary inside the detector-facing sector. This complete capture has a simple origin rather than being a claim about a real detector: at $D = 0.55$ the orthogonal

edge $q_{\text{edge}}^{(y)} = 0.737923$ lies above the scanned interval $q_{\text{ref}} \leq 0.730$ ($q \leq 0.7315$ for 500 Th after mass scaling), so the Y mode of the prescribed 10^{-4} ensemble never becomes unstable during the scan and every first crossing occurs within a small fraction of a degree of the detector axis. The same ensemble at equal duty loses 32–36% of first crossings to off-axis branches (Table 2), so the difference is produced by the edge split, not by the cold ensemble; for the broad diagnostic ensemble of Fig. 4(a)–(c), whose scan also passes the orthogonal edge, a residual off-axis fraction of order 10% remains. The off-axis loss of a realistic packet is therefore governed by its Y covariance at launch and by how far the scan penetrates beyond $q_{\text{edge}}^{(y)}$. This remains an ideal transverse calculation with the same exclusions as Fig. 3; unequal duty is the preferred baseline axis-selectivity refinement for subsequent three-dimensional optimization.⁹ Table 5 distinguishes the candidate modes.

Table 4: Outcome counts and accepted-event statistics from the recalculated preferred-baseline benchmark of Fig. 4(d): $D = 0.55$, $q_d = 0.10$ for the 501-Th reference mass, $N_{\text{launch}} = 1000$ per mass, the same cooled covariance and scan range as Table 2, and the independently determined calibration $q_e = 0.705869$. The reported standard deviations describe the numerical ensemble only and are not instrumental resolving powers.

True m/z	N_{det}	N_{other}	N_{opp}	$N_{\text{uncrossed}}$	Mean $\widehat{m/z}$	SD (Th)
500	1000	0	0	0	499.891	0.183
501	1000	0	0	0	501.000	0.183
502	1000	0	0	0	501.902	0.184

Table 5: Qualitative comparison of product-ion ejection modes.

Mechanism	Physical role	Status
RF-only instability	Mass-dependent boundary loss without directional bias	Limiting case
Equal-duty DC $n = 1$ plus $a = 0$ RF scan	Detector-side symmetry breaking with degenerate transverse edges	Baseline reference
Unequal-duty DC $n = 1$ plus $a = 0$ RF scan	Detector-directed instability with the orthogonal mode retained	Preferred baseline
Standing AC $n = 1$	Fixed-axis secular resonance ejection inside the stable region	Supplement S2
Rotating $n = 1$	Circular-mode excitation with a continuously changing direction	Supplement S2

6 Fragment Catch, Gas Cycle, and Timing

A pure RF quadrupole has the usual low-mass cutoff. If a precursor of mass m_{prec} and charge-number magnitude z_{prec} is held at q_{prec} , a fragment of mass m_f and charge-number magnitude

⁹The required field and scan validation sequence is given in Supplement S6.

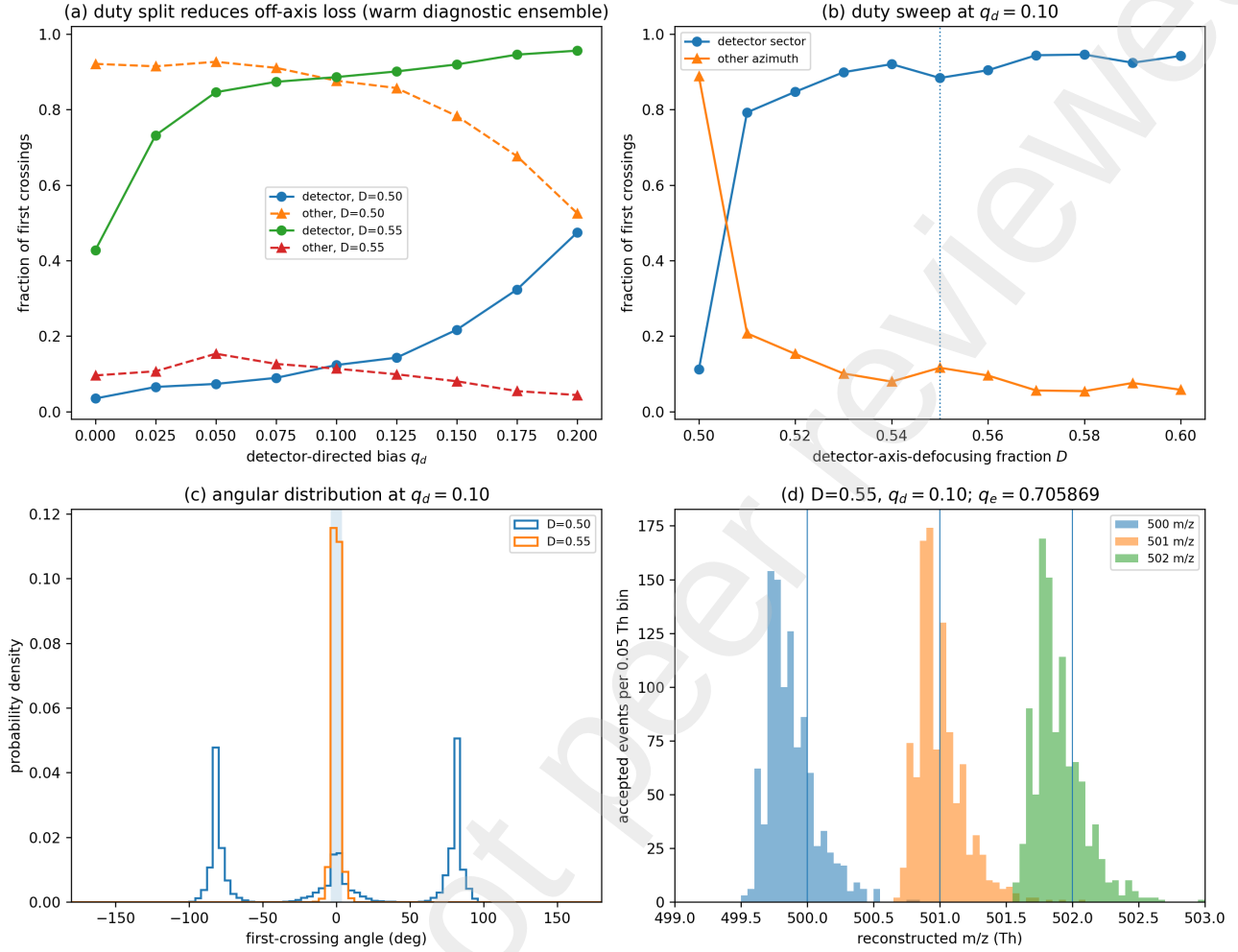


Figure 4: Recalculated RF duty-cycle dependence of the reduced 24-blade model. Here D is plotted directly as the fraction of the period during which the detector-axis coordinate is defocused. (a) First-crossing fractions versus detector-directed bias for $m/z = 501$ at $D = 0.50$ and 0.55 using a deliberately broad diagnostic ensemble. (b) Detector-sector and other-azimuth fractions versus D at fixed $q_d = 0.10$. The recalculation shows a broad high-selection region rather than a unique narrow optimum; the vertical line marks the baseline $D = 0.55$. (c) Angular distributions at $D = 0.50$ and 0.55 for the same diagnostic ensemble. Panels (a)–(c) use $q_{\text{ref}} = 0.620 \rightarrow 0.820$ over 500 RF periods with normalized diagonal Gaussian widths $\sigma_X = \sigma_{X'} = \sigma_Y = \sigma_{Y'} = 0.02$; panel (a) uses 2000 ions per bias value and panels (b,c) 6000 ions per duty value. (d) Accepted-event mass spectrum for the cooled benchmark at $D = 0.55$, $q_d = 0.10$, using 1000 ions per mass and an independently determined 501-Th calibration $q_e = 0.705869$. The same q_e is then applied unchanged to 500 and 502 Th. No instrumental resolving power is inferred. All panels use the ideal circular-boundary model and exclude the three-dimensional aperture field, finite blade length, collisions, space charge, end fields, and collector optics.

z_f experiences

$$q_f = q_{\text{prec}} \frac{z_f}{z_{\text{prec}}} \frac{m_{\text{prec}}}{m_f}.$$

For equal precursor and fragment charge states, $z_f = z_{\text{prec}}$, the retention condition $q_f < q_b$ becomes

$$\frac{m_f}{m_{\text{prec}}} > \frac{q_{\text{prec}}}{q_b}.$$

Thus, for $q_{\text{prec}} = 0.3$ and equal charge states, fragments below approximately $0.42m_{\text{prec}}$ cannot be retained at the same quadrupole setting.

For ideal angular order n , use

$$\eta_n(\rho) = \frac{n(n-1)}{2} q_n \left(\frac{\rho}{R} \right)^{n-2}, \quad q_n = \frac{4V_n}{\mu R^2 \Omega^2},$$

with $\eta_n \lesssim 0.3$ as a conservative condition [36]. At $q_6 = 0.3$, the resulting adiabatic radius is about $0.51R$. The $n = 6$ channel therefore supplies a useful outer catch barrier while retaining a larger practical well than $n = 12$.

A pure $n = 6$ field has vanishing harmonic curvature at the axis. A complete $n = 2 \rightarrow n = 6$ handoff therefore does not automatically preserve small-amplitude transverse action. The baseline retains a residual quadrupole during activation or requires direct ensemble validation. After activation, full $n = 2$ confinement is restored while gas remains available for recompression. The packet is then held in the narrowed $n = 2$ well for several damping times, the detector-directed DC field is introduced gradually, and only then is the gas pumped out for the resolving scan. This ordering minimizes coherent breathing and establishes the phase-space distribution required by the ejection calculation.

During the baseline activation stage, the precursor remains trapped while the standing AC $n = 1$ field drives transverse secular motion in the gas-loaded hybrid catch. The instantaneous collision rate can therefore be written

$$\nu_{\text{col}}(t) = \frac{p(t)}{k_B T_g} \langle \sigma_{\text{col}}(v_{\text{rel}}) v_{\text{rel}} \rangle, \quad N_{\text{col}} = \int_0^{t_{\text{act}}} \nu_{\text{col}}(t) dt,$$

where v_{rel} is the ion–gas relative speed and the brackets denote the appropriate velocity average. For approximately constant pressure and rate coefficient during the activation interval,

$$N_{\text{col}} \simeq \frac{p t_{\text{act}}}{k_B T_g} \langle \sigma_{\text{col}}(v_{\text{rel}}) v_{\text{rel}} \rangle.$$

No prescribed single-pass path length appears: the resonantly driven ion samples the gas throughout its trapped trajectory. Because the effective rate coefficient changes with collision gas, ion energy, resonant amplitude, and damping, the present design does not assign a universal collision count or CID pressure. Fragment yield must instead be optimized jointly with V_{act} , $\omega_{\text{act}}(t)$, t_{act} , pressure, and the hybrid-catch composition. For

$$p(t) = p_0 e^{-t/\tau_p}, \quad \tau_p = \frac{V_{\text{gas}}}{S_{\text{eff}}},$$

the illustrative values $V_{\text{gas}} = 15 \text{ cm}^3$ and $S_{\text{eff}} = 5 \text{ L s}^{-1}$ give $\tau_p \simeq 3 \text{ ms}$. Pumping from 10 to 0.03 mTorr takes approximately 17 ms in the conductance-free model. The 10 mTorr starting

value is an illustrative pressure-recovery excursion, not a CID optimum inferred from the collision model. The calculated pump time is an ideal lower bound, not a prediction for a closed cylindrical housing. The realised instrument must preserve substantial radial gas conductance through a grounded mesh or slotted shield; a close-fitting solid shield would invalidate this estimate by forcing the gas predominantly through the axial path.¹⁰

For singly charged ions,

$$N_{\text{acc}} = \frac{I_{\text{ion}} T_{\text{acc}}}{e} \simeq 3.1 \times 10^5 \left(\frac{I_{\text{ion}}}{1 \text{ pA}} \right) \left(\frac{T_{\text{acc}}}{50 \text{ ms}} \right).$$

Direct simulations of multispecies ion-ion coupling in quadrupole ion traps report signal fusion, suppression, distortion, and frequency shifts at modest populations [37]. Space-charge shifts, rather than geometric capacity, are expected to limit the useful packet.

Table 6 gives an order-of-magnitude timing budget.

Table 6: Order-of-magnitude timing budget for one analytical packet.

Stage	Aggressive	Baseline	Conservative
Injection and capture	0.1–0.5 ms	0.5–2 ms	2–5 ms
Precursor isolation	0.5–2 ms	2–5 ms	5–15 ms
Cross-fade to catch field	0.02–0.05 ms	0.05–0.20 ms	0.2–1.0 ms
Gas rise and resonant activation/catch	0.7–2 ms	2–8 ms	8–30 ms
Restore $n = 2$ and recompress	0.5–1 ms	1–3 ms	3–10 ms
Pump-out to resolving pressure	5–10 ms	15–30 ms	40–80 ms
RF scan and DC-biased boundary ejection	1–3 ms	3–10 ms	10–30 ms
Total	8–19 ms	24–58 ms	68–171 ms

A baseline analytical time of about 24–58 ms, combined with $T_{\text{acc}} \simeq 50$ ms, gives a packet rate of order 20 Hz.

Supplementary Table S3 (Supplement S10) collects, in one place, the baseline specifications that are otherwise distributed over Sections 3–6 and Supplements S4, S5, S7, and S8; it is intended as a single reference for an experimental implementation.

The reduced transverse calculation establishes only the directional effect of the DC $n = 1$ bias, the first-crossing categories, and relative changes between equal- and unequal-duty cases.¹¹ No instrumental resolving power or absolute transmission is inferred from the reduced model.

7 Conclusion

This work proposes a single 24-blade cylindrical analyser that performs the principal functions of tandem mass spectrometry sequentially in the same analytical volume. Precursor isolation, activation, fragment retention, recompression, pressure recovery, and product-ion analysis are obtained by reprogramming the transverse boundary rather than by transferring ions through

¹⁰The collision-count and pump-out estimates are detailed in Supplement S4.

¹¹Event reconstruction, raw event reporting, and the detailed validation protocol are collected in Supplement S7.

separate Q1, q2, and Q3 analysers. The baseline operating sequence uses an $n = 2$ quadrupole for precursor selection and final mass analysis, a predominantly $n = 6$ field with residual $n = 2$ curvature as the fragment catch, a standing transverse AC $n = 1$ field for resonant collisional activation, and a detector-directed DC $n = 1$ component during product-ion ejection. Conventional end electrodes provide axial confinement and transfer but are not used as the baseline CID excitation mechanism.

An important feature of the proposed cycle is that the fragment-catch field is already present while the precursor is activated. After activation, the $n = 2$ quadrupole is restored while damping gas remains in the cage so that the fragment packet can recompress and rethermalize before pump-out. The resolving stage then begins at low pressure, with the quadrupolar DC parameter set to $a = 0$. In this way the same electrode array changes function in time without requiring physical transfer of the product-ion packet between successive mass-analysis regions.

For product-ion readout, the preferred baseline is DC-biased stability-boundary ejection with unequal RF duty. The square-wave $n = 2$ RF amplitude is scanned and provides the mass coordinate, while a detector-directed DC $n = 1$ field remains continuously superimposed. With D defined as the detector-axis-defocusing fraction, $D \simeq 0.55$ separates the two transverse instability boundaries so that the detector-axis mode becomes unstable first. The DC field then selects the detector-facing branch of that instability. The reduced calculations show that opposite-side escape is not intrinsic to stability-boundary ejection and can be suppressed in the ideal transverse model. No rotating field, auxiliary resonant excitation during the product scan, or final extraction pulse is required for the baseline readout.

The numerical first-crossing calculations should nevertheless be interpreted as validation of the operating principle rather than as predictions of instrumental resolving power or absolute transmission. The calculated non-detector-sector fraction depends strongly on the post-cooling phase-space covariance, and the present circular-boundary model does not determine whether such trajectories strike a blade, pass through another gap, or are recovered by the three-dimensional extraction field. Final analytical performance will therefore depend on the collisional $n = 6 \rightarrow n = 2$ transfer, the three-dimensional detector aperture and collection optics, the RF scan law, DC steering, space charge, gas history, and the measured electrode waveforms. Higher odd DC orders may provide additional near-wall steering, but that possibility remains to be demonstrated.

The programmable operation is enabled by the underlying cylindrical boundary synthesis. In the present formulation the transverse harmonics and their orientations are represented in real $\text{Cl}(1, 3)$: the bivector σ_{12} supplies the in-plane phase structure, electric fields are relative vectors in $\text{Cl}^+(1, 3)$, and the independently driven blades realize the required $n = 1$, $n = 2$, $n = 6$, and other angular coefficients. The Clifford formulation is therefore used here primarily as a compact synthesis and bookkeeping framework for a physically programmable electrode array; the central instrumental proposal is the time-ordered use of those synthesized fields in a single tandem-in-time analyser.

The most important next steps are experimental and engineering ones: collisional simulation and measurement of the $n = 6 \rightarrow n = 2$ recompression, calibration of the dynamic stability-boundary scan, optimization of the duty cycle and detector-directed DC field, a complete three-dimensional model of the inter-blade aperture and collector, and trajectory calculations through the full activation, cooling, pump-out, and analysis sequence. Prospective routes for improving resolution, transmission, throughput, and calibration stability are summarized in Supplement S9. These tests will determine whether the analytical advantages suggested by the

single-cage architecture can be realized in a practical instrument.

Acknowledgments

The author gratefully acknowledges Ilia Baranov of Polymath Robotics for fruitful discussions and practical suggestions concerning the design of the drive electronics and the calibrated implementation of the programmable boundary.

Notation

Symbol	Meaning
$\text{Cl}(1, 3)$	Real spacetime Clifford algebra, signature $(+ - - -)$
e, e, e_μ	Exponential function (roman), elementary charge, and spacetime basis vectors, respectively
$e_\mu, \eta_{\mu\nu}$	Spacetime basis and Minkowski metric
$\sigma_k = e_k e_0$	Relative spatial vectors in $\text{Cl}^+(1, 3)$
σ_{12}	Transverse bivector and geometric phase generator
$\langle \cdot \rangle_0, \sim$	Scalar-grade projection and Clifford reversion
$\mathbf{r}_\perp, \zeta$	Transverse relative vector and Clifford coordinate
$\mathcal{C}_\perp$	$\text{span}_{\mathbb{R}}\{1, \sigma_{12}\}$
ρ, θ, s	Radial, azimuthal, and axial coordinates
R	Birdcage reference radius
g, ψ_n	Boundary schedule and order- n Clifford channel
$\hat{\psi}_n(k, t)$	Axial Clifford-Fourier coefficient
A_n, φ_n	Channel amplitude and orientation
I_n	Modified Bessel function of the first kind
$\theta_k, \Delta\theta_b$	Blade-centre angle and blade aperture
d_b	Blade fill factor, $\Delta\theta_b/(2\pi/24)$
$\theta_g, \Delta\theta_g$	Detector-gap angle and full gap width
W_n	Finite-width pre-emphasis factor
κ_n	Geometric coupling, realized order- n coefficient per unit peak blade voltage ($\kappa_2 = 0.869$)
V_n^{blade}	Peak blade voltage commanded for realized coefficient V_n
U, V_{rf}	Quadrupole DC and RF coefficients
a, q, q_n	Dimensionless stability parameters
q_b, q_e	Ideal first RF boundary and calibrated ejection coordinate
$Q = \pm ze, z, \mu, (m/z)_{\text{Th}}$	Ion charge, charge-number magnitude, and mass-to-charge values
Ω, f_{RF}	Angular and ordinary RF frequencies
$M(q)$	One-period monodromy matrix
β, ω_0	Secular ratio and secular angular frequency
V_d, q_d	Detector-directed DC dipole voltage and dimensionless bias
D	RF duty: detector-axis-defocusing fraction of one period
$\kappa = V_d/V_{\text{rf}}$	Optional proportional DC-bias ratio for a broad scan
$\mathbf{b}(q), \mathbf{w}, \mathbf{w}_c$	Affine forcing vector, detector-axis state, and forced periodic state
$T_\perp, \sigma_x, \sigma_y$	Effective transverse temperature and cooled spatial widths
ω_x, ω_y	Transverse secular angular frequencies
x_g, y_g	Detector-directed and orthogonal transverse coordinates
ϵ_2	Residual quadrupole amplitude fraction during fragment capture
$V_{\text{act}}, \omega_{\text{act}}, \phi_{\text{act}}, \theta_{\text{act}}, t_{\text{act}}$	AC dipole activation amplitude, angular frequency, phase, orientation, and duration
V_{2p+1}	Odd-order DC steering coefficients
η_n	High-order adiabaticity parameter
$T_{\text{packet}}, T_{\text{acc}}, T_{\text{analytical}}$	Packet period, accumulation time, and analytical time
$\chi, \tilde{\chi}_D$	Unit square-wave drive and zero-mean unequal-duty waveform
$a_{\text{tip}}, q_{\text{tip}}$	Equal-duty stability-tip constants

Symbol	Meaning
α	Square-wave Hill parameter, $\alpha = (\pi/2)\sqrt{2q}$
A_x	Detector-axis oscillation amplitude about the forced orbit
$T(\nu, L)$	Constant-segment transfer matrix (Appendix C)
$\sigma_{\text{col}}(v_{\text{rel}}), v_{\text{rel}}, N_{\text{col}}$	Collision cross-section, ion–gas relative speed, and mean collision number
$p, T_g, V_{\text{gas}}, S_{\text{eff}}, \tau_p$	Gas pressure, temperature, volume, pumping speed, and time constant
$N_{\text{det}}, N_{\text{other}}, N_{\text{opp}}, N_{\text{uncrossed}}$	First-crossing outcome counts (Supplement S7)
$\mathcal{D}_{501}$	Accepted detector-sector event set of the reference mass
$\lambda_k, \hat{c}_n, C_{\text{eff}}$	Passive divider ratio, modal capacitance, and effective load

Appendix

A Transverse Clifford Multipole Identity

Let

$$\zeta = x + y\sigma_{12}, \quad \tilde{\zeta} = x - y\sigma_{12}, \quad \nabla_{\perp} = \sigma_1\partial_x + \sigma_2\partial_y.$$

For every integer $n \geq 1$,

$$\nabla_{\perp} \langle \zeta^n \rangle_0 = n\sigma_1 \tilde{\zeta}^{n-1}.$$

Hence

$$\Phi_n = \frac{V_n}{R^n} \langle \zeta^n \rangle_0, \quad \mathcal{E}_n = -\nabla_{\perp} \Phi_n = -\frac{nV_n}{R^n} \sigma_1 \tilde{\zeta}^{n-1} \in \text{Cl}^+(1, 3).$$

The first relevant scalar harmonics are

$$\begin{aligned} \langle \zeta \rangle_0 &= x, & \langle \zeta^2 \rangle_0 &= x^2 - y^2, \\ \langle \zeta^6 \rangle_0 &= x^6 - 15x^4y^2 + 15x^2y^4 - y^6, & \langle \zeta^{12} \rangle_0 &= \rho^{12} \cos 12\theta. \end{aligned}$$

B Cylindrical Bessel Transfer Kernel

In a source-free cylinder,

$$\frac{1}{\rho} \partial_{\rho}(\rho \partial_{\rho} \Phi) + \frac{1}{\rho^2} \partial_{\theta}^2 \Phi + \partial_s^2 \Phi = 0.$$

For angular order n and axial wavenumber k , regularity on the axis selects $I_n(|k|\rho)$. Boundary normalization at $\rho = R$ gives

$$F_{n,k}(\rho) = \frac{I_n(|k|\rho)}{I_n(|k|R)},$$

which yields Eq. (5). For $k \rightarrow 0$,

$$F_{n,k}(\rho) \longrightarrow (\rho/R)^n.$$

C Compact Equal-Duty Monodromy Formula

The Hill coordinate u and its derivative are real scalars. The following 2×2 transfer matrices act on the real state $(u, u')^T$; they are numerical operators over $\mathbb{R}$, not additional phase variables. For a constant segment $u'' + \nu u = 0$ of dimensionless length L , define

$$T(\nu, L) = \begin{pmatrix} C(\nu, L) & S(\nu, L) \\ -\nu S(\nu, L) & C(\nu, L) \end{pmatrix}, \quad \det T = 1,$$

where

$$C(\nu, L) = \begin{cases} \cos(\sqrt{\nu}L), & \nu > 0, \\ 1, & \nu = 0, \\ \cosh(\sqrt{-\nu}L), & \nu < 0, \end{cases}$$

and

$$S(\nu, L) = \begin{cases} \sin(\sqrt{\nu}L)/\sqrt{\nu}, & \nu > 0, \\ L, & \nu = 0, \\ \sinh(\sqrt{-\nu}L)/\sqrt{-\nu}, & \nu < 0. \end{cases}$$

For the equal-duty square wave,

$$M_x = T(a + 2q, \pi/2)T(a - 2q, \pi/2), \quad M_y = T(-a - 2q, \pi/2)T(-a + 2q, \pi/2),$$

and stability requires $|\text{Tr } M_x| \leq 2$ and $|\text{Tr } M_y| \leq 2$.

For the detector-axis equation with a constant segment force,

$$u'' + \nu u = f,$$

the affine segment map is

$$\begin{pmatrix} u \\ u' \end{pmatrix}_L = T(\nu, L) \begin{pmatrix} u \\ u' \end{pmatrix}_0 + f \begin{pmatrix} [1 - C(\nu, L)]/\nu \\ S(\nu, L) \end{pmatrix},$$

with the continuous $\nu \rightarrow 0$ limit $(fL^2/2, fL)^\top$. Composing the two square-wave half-cycles gives the affine monodromy of Eq. (8). The stability criterion remains determined solely by the homogeneous matrix $M(q)$.

D Secular Pseudopotential Used in the Gap-Axis Plot

Represent the laboratory electric field as the relative vector

$$\mathcal{E}(\mathbf{r}, t) = \sum_{k=1}^3 E^k(\mathbf{r}, t) \sigma_k = \mathcal{E}_0(\mathbf{r}) \chi(t) \in \text{Cl}^+(1, 3).$$

Because $\sigma_k^2 = 1$, its Euclidean magnitude is the scalar

$$|\mathcal{E}_0|^2 \equiv \langle \mathcal{E}_0^2 \rangle_0.$$

The small-micromotion pseudopotential is therefore

$$U_{\text{ps}}(\mathbf{r}) = C_\chi \frac{(ze)^2}{m\Omega^2} \langle \mathcal{E}_0^2(\mathbf{r}) \rangle_0.$$

For an equal-duty bipolar square wave specified by its peak amplitude,

$$C_\chi = \frac{4}{\pi^2} \sum_{r \text{ odd}} \frac{1}{r^4} = \frac{\pi^2}{24}.$$

Only relative potential is plotted in Fig. 2, so the common dimensional factor is suppressed. Near the first instability boundary, where micromotion is large and the secular approximation loses quantitative control, this pseudopotential is used only as a qualitative visualization of detector-side displacement. All ejection and calibration claims in the main text are based on the exact time-dependent Hill equations and affine monodromy, not on this effective potential. For an ideal angular order n ,

$$\frac{U_{\text{ps},n}(\rho)}{U_{\text{ps},n}(R)} = \left(\frac{\rho}{R} \right)^{2n-2}.$$

Supplementary material

Supplementary material S1–S9 (alias protection and blade fill factor; AC-dipole activation and alternative readouts; sequential Q1/q2/Q3 benchmark; gas and pump-out calculations; voltage convention; field and scan validation sequence; event reconstruction and reporting; electronics and calibrated boundary implementation; prospective routes to higher performance; baseline specification summary), including Fig. S1 and Tables S1–S3 (S10: baseline specification summary), is provided as a separate file with this article.

Declarations

CRedit authorship contribution statement. V. Baranov: conceptualization, methodology, formal analysis, software, visualization, writing – original draft, writing – review and editing.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Overleaf.com, a LaTeX editor with integrated AI features, Claude (Anthropic), and ChatGPT (OpenAI) for assistance with Python and Mathematica code, numerical cross-checks, consistency checking, and improvement of language and readability. The author reviewed, verified, and edited all AI-assisted output as needed and takes full responsibility for the content of the publication.

Data availability. No experimental data were generated. The Python recalculation script, raw first-crossing event tables, and numerical summary files underlying Table 2 and Figs. 3 and 4, together with the boundary-element and Mathematica outputs underlying Fig. S1 and Table S2, are available upon request

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